

ChainBenders

TalentLens AI

AI-Powered Fair Recruitment Platform

Resume Parsing · Bias Detection · Explainable AI · Blockchain Verification

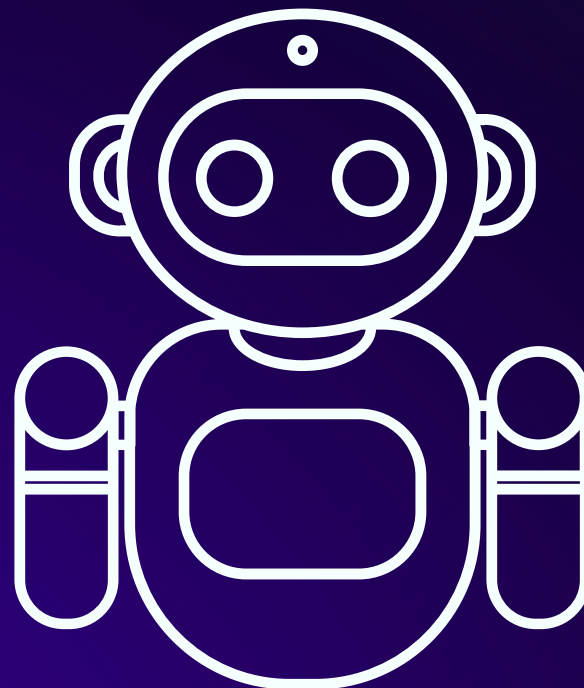
Technologies

→ AI & Machine Learning

→ FastAPI + React

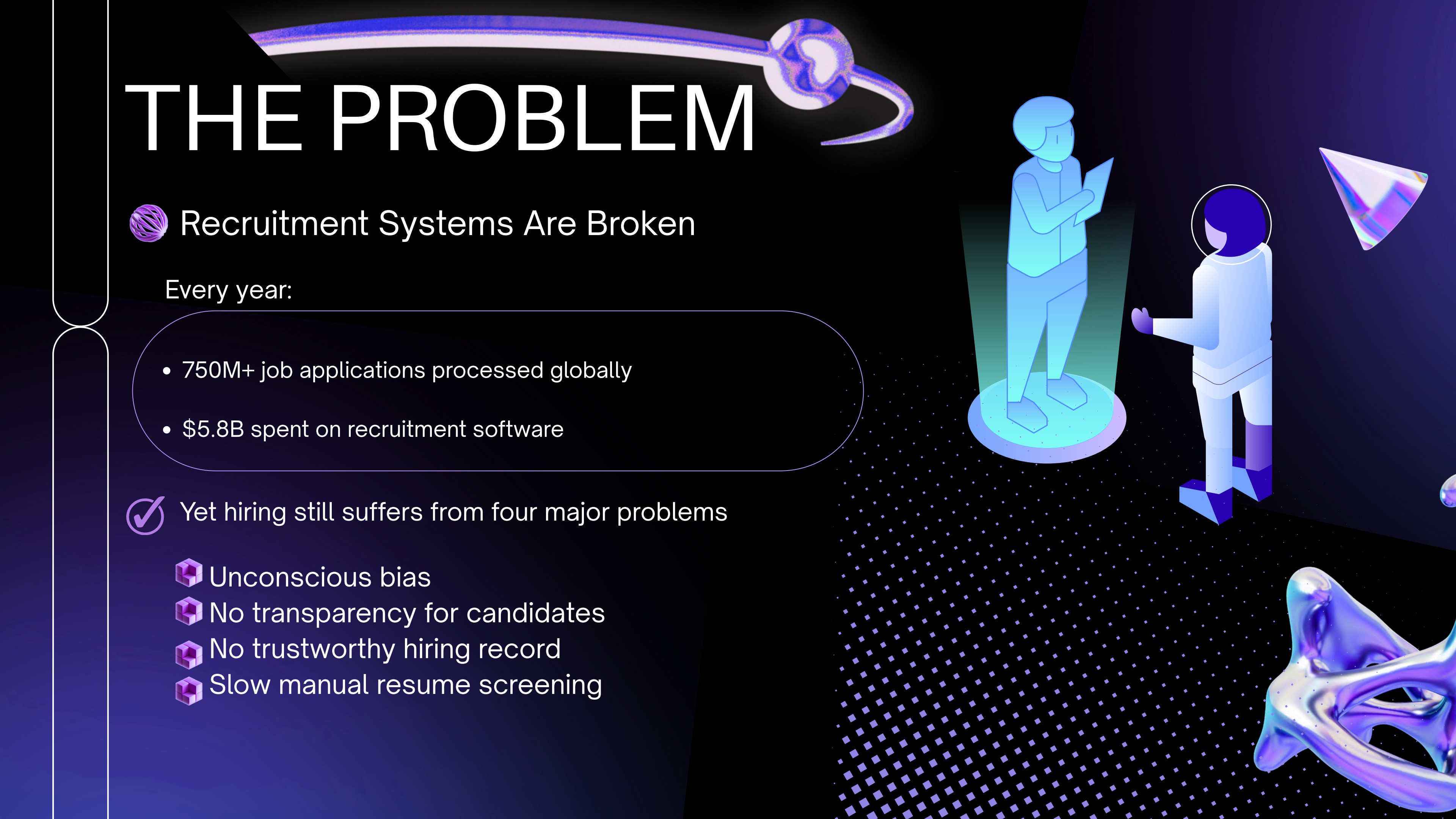
→ Sentence-BERT NLP

→ Polygon Blockchain



City Fusion X Hackathon 2026

THE PROBLEM



🌀 Recruitment Systems Are Broken

Every year:

- 750M+ job applications processed globally
- \$5.8B spent on recruitment software

✅ Yet hiring still suffers from four major problems

- 📦 Unconscious bias
- 📦 No transparency for candidates
- 📦 No trustworthy hiring record
- 📦 Slow manual resume screening



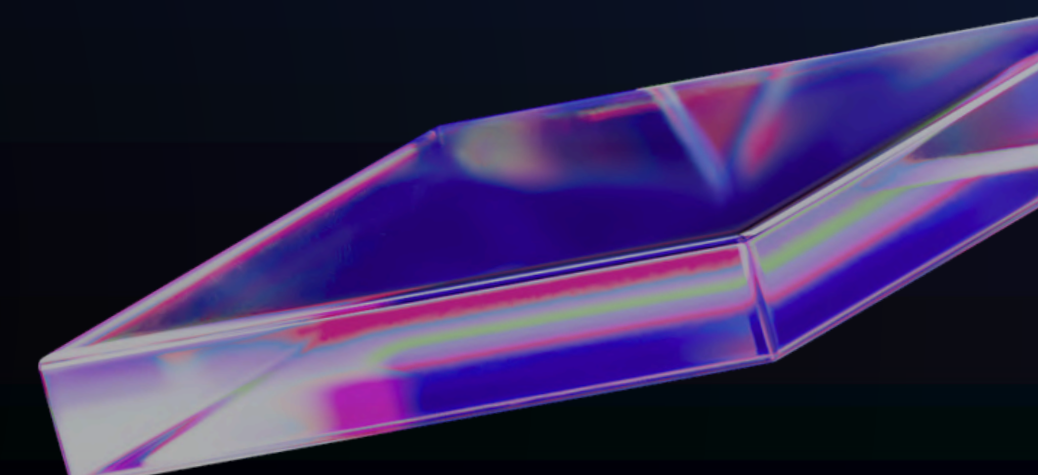
Industry Impact

Real Consequences

Metric	Reality
Qualified candidates rejected due to bias	47%
Candidates receiving no feedback	78%
Time to manually screen 100 resumes	40–60 hours
Cost of bad hire	\$15k–\$50k
Companies without hiring audit trail	63%



Current ATS systems automate the problem instead of solving it.



Introducing TalentLens AI

TalentLens AI is an AI-powered recruitment platform that ensures:

- ✓ Fair hiring
- ✓ Transparent decisions
- ✓ Tamper-proof records
- ✓ Faster recruitment



Core Technologies

- Artificial Intelligence
- Machine Learning
- Explainable AI
- Blockchain Verification



Platform Overview

TalentLens AI System

- TalentLens combines three innovation pillars

Pillar

- Fair Screening
- Transparent Decisions
- Trustworthy Records

Solution

- AI evaluates only skills and experience
- Explainable AI reports for every candidate
- Hiring decisions stored on blockchain

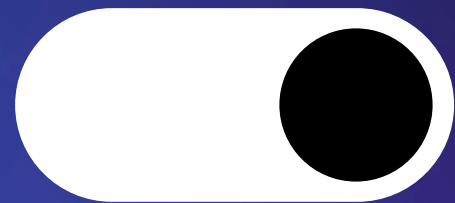
Result:

- Fair candidate ranking
- Full decision transparency
- Immutable audit trail

Key Features

TalentLens AI Capabilities

- AI Resume Parsing
- Semantic Skill Matching
- Bias Detection Engine
- Explainable AI Reports
- Blockchain Verification
- Recruiter Analytics Dashboard
- AI Job Description Generator



AI Resume Parsing

Automated Resume Analysis

Supported formats

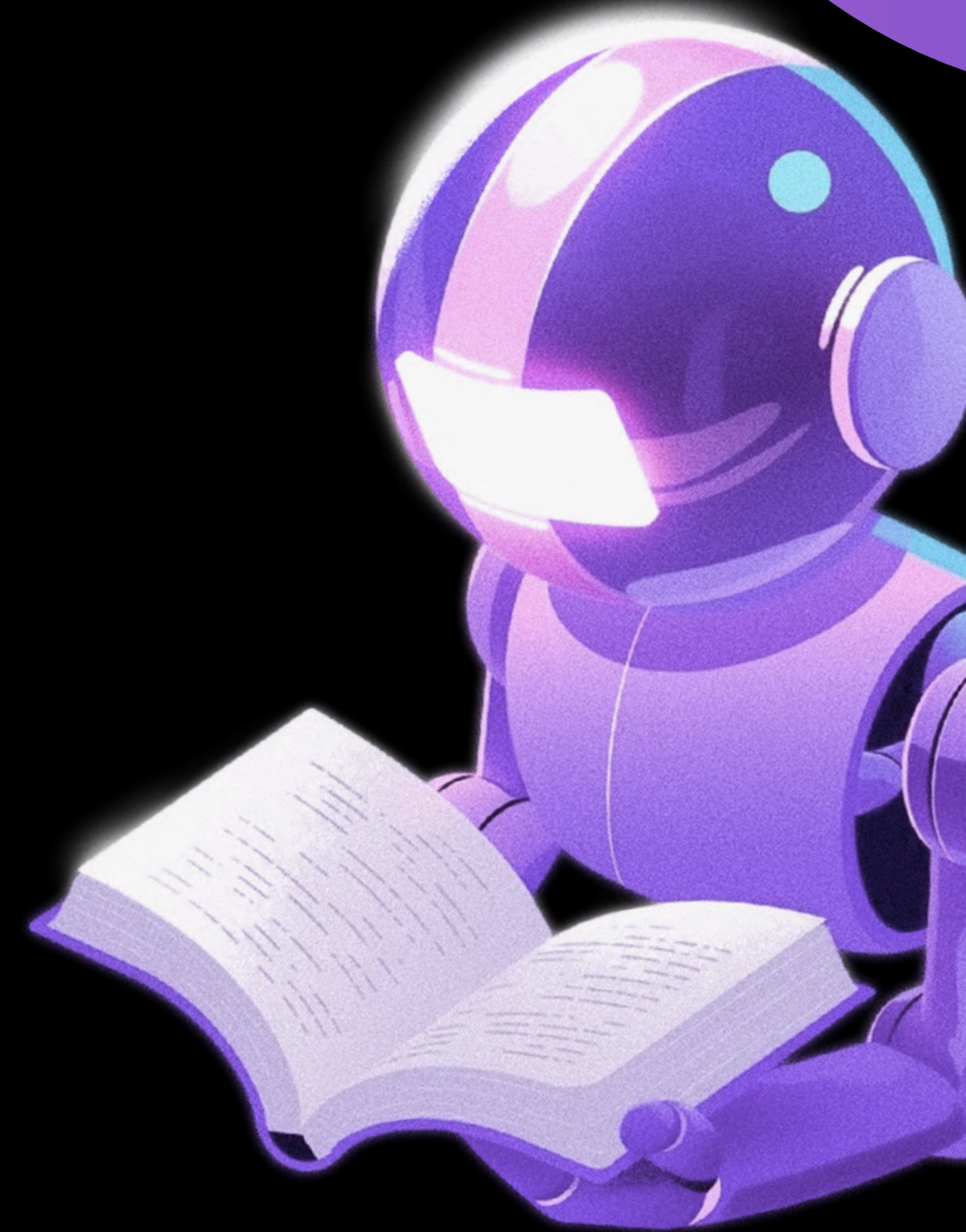
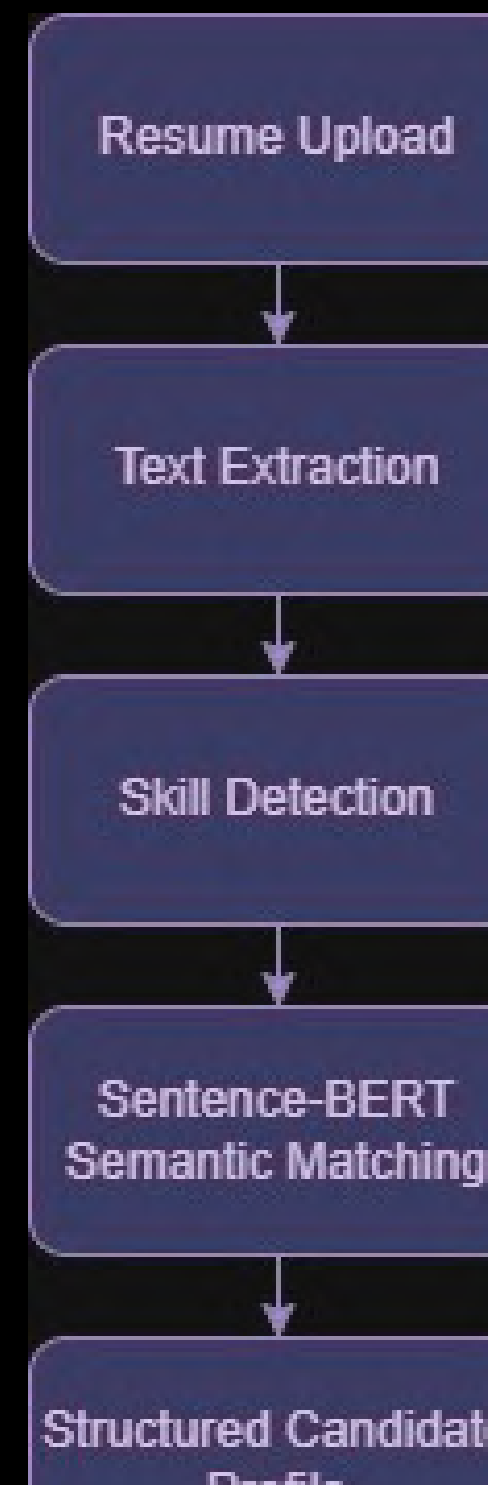
- PDF
- DOCX



Example output

- Skills: Python, React, Docker
- Experience: 4.5 years
- Projects: ML Fraud Detection System

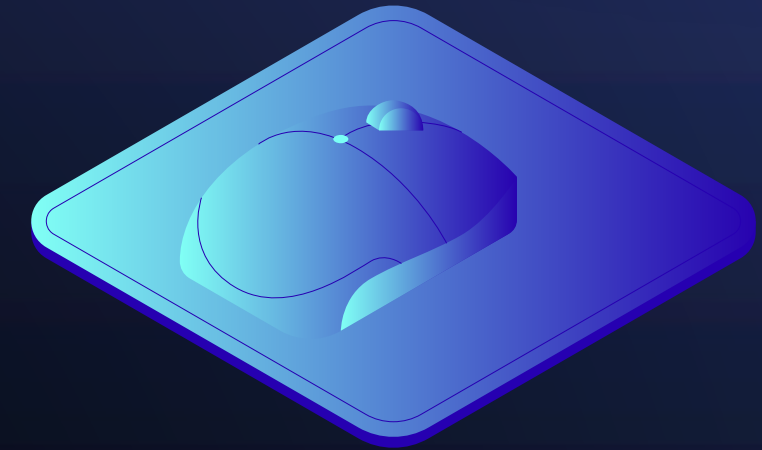
Processing pipeline





Multi-Factor Scoring Model

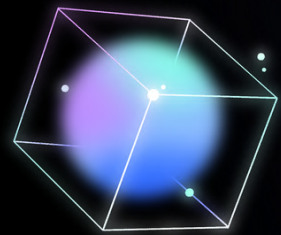
AI Candidate Ranking



Factor	Weight
Skill Match	40%
Experience	25%
Projects	20%
Skill Diversity	10%
Soft Skills	5%

Final Score Formula

$\text{FinalScore} = \text{RawScore} \times \text{FairnessFactor}$
Ensures bias reduces ranking automatically.



Bias Detection System

TalentLens detects three bias types



Bias

Experience Bias
Education Bias
Career Gap Bias

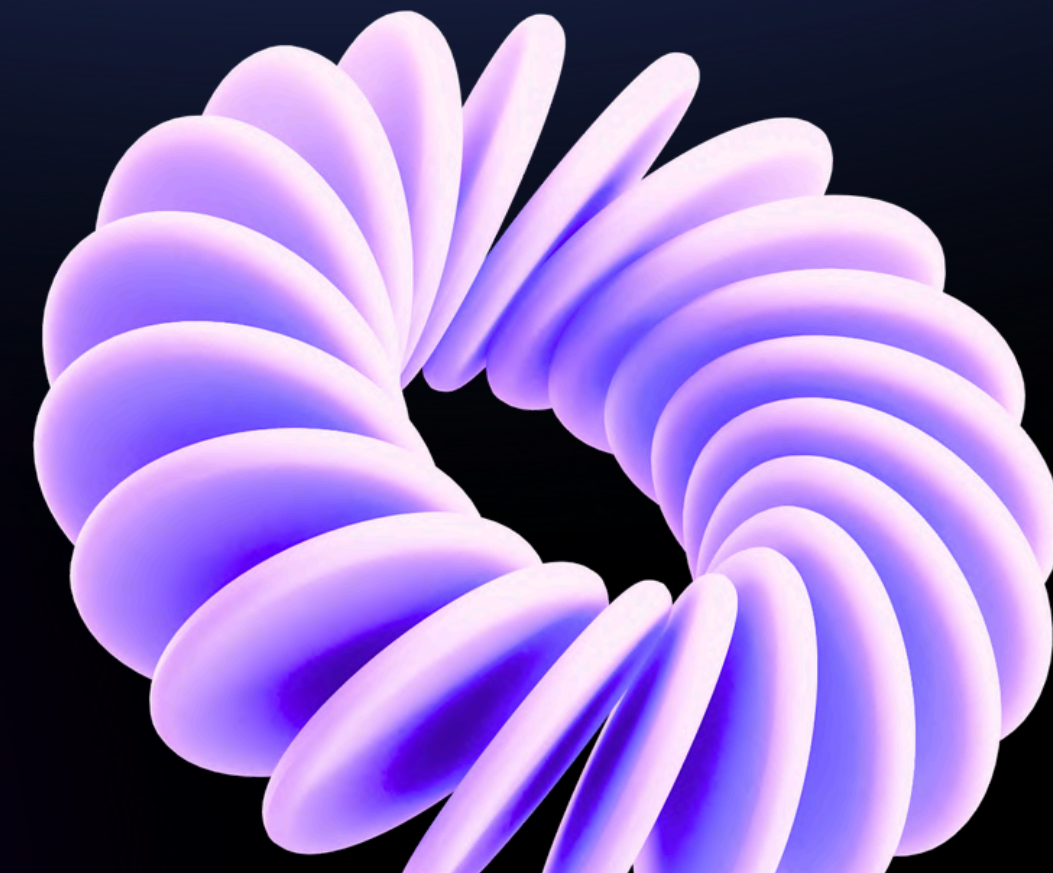


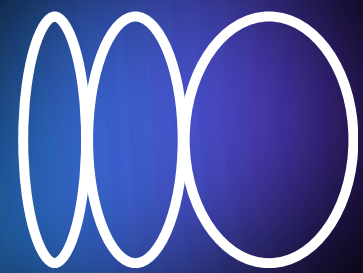
Purpose

Prevents junior candidate discrimination
Avoids elite-school preference
Fair evaluation for career breaks

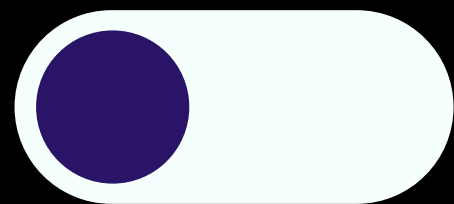


- ▲ Fairness Score
- ▲ Range: 0–100
- ▲ Score above 90 = fair evaluation





Explainable AI



Transparent Hiring Decisions

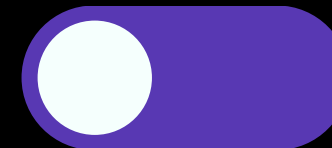
Instead of a black box decision, TalentLens provides:



Candidate report includes:

- ✓ Score breakdown
- ✓ Strengths
- ✓ Concerns
- ✓ Improvement suggestions

↗ Example



- Strength: Strong Python & React skills
- Concern: Limited DevOps experience
- Supports GDPR explainable AI compliance



Blockchain Verification



Tamper-Proof Hiring Records

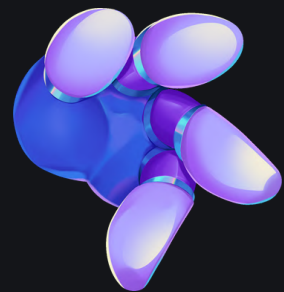
TalentLens stores hiring decisions on the Polygon blockchain.

Each record includes

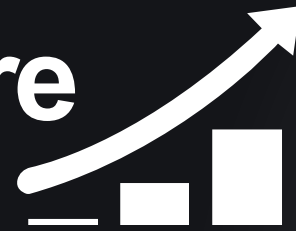
- ➔ Resume hash
- ➔ AI score
- ➔ Fairness score
- ➔ Decision
- ➔ Timestamp

Benefits

- ✓ Immutable records
- ✓ Transparent audits
- ✓ Trustworthy hiring



System Architecture



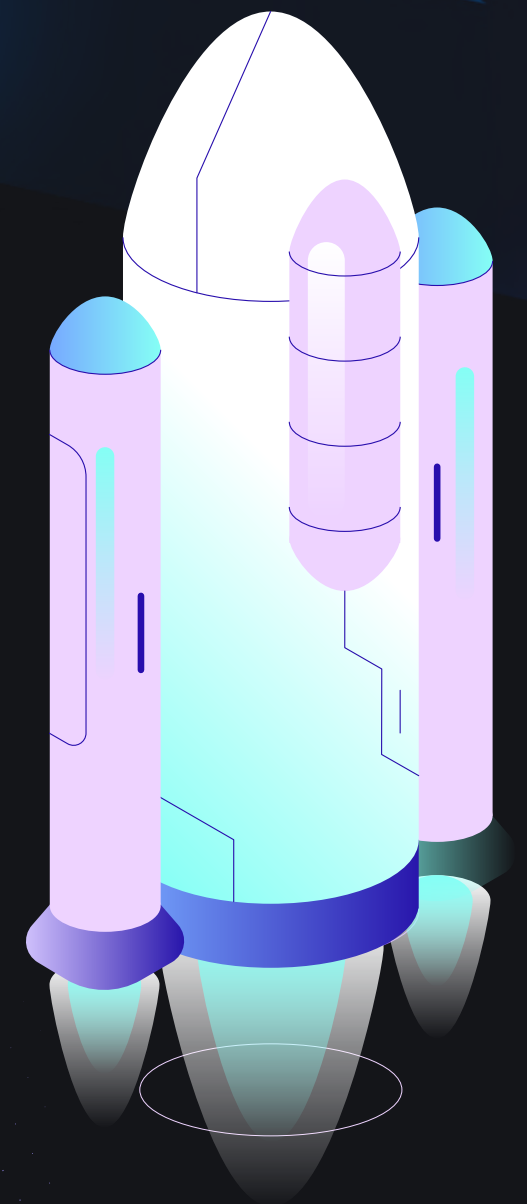
Technology Stack



Layer	Technology
Frontend	React + TailwindCSS
Backend	FastAPI
AI/NLP	spaCy + Sentence-Transformers
Database	MySQL / SQLite
Blockchain	Solidity + Polygon
Authentication	JWT

Deployment plan

- Backend → Heroku
- Frontend → Vercel

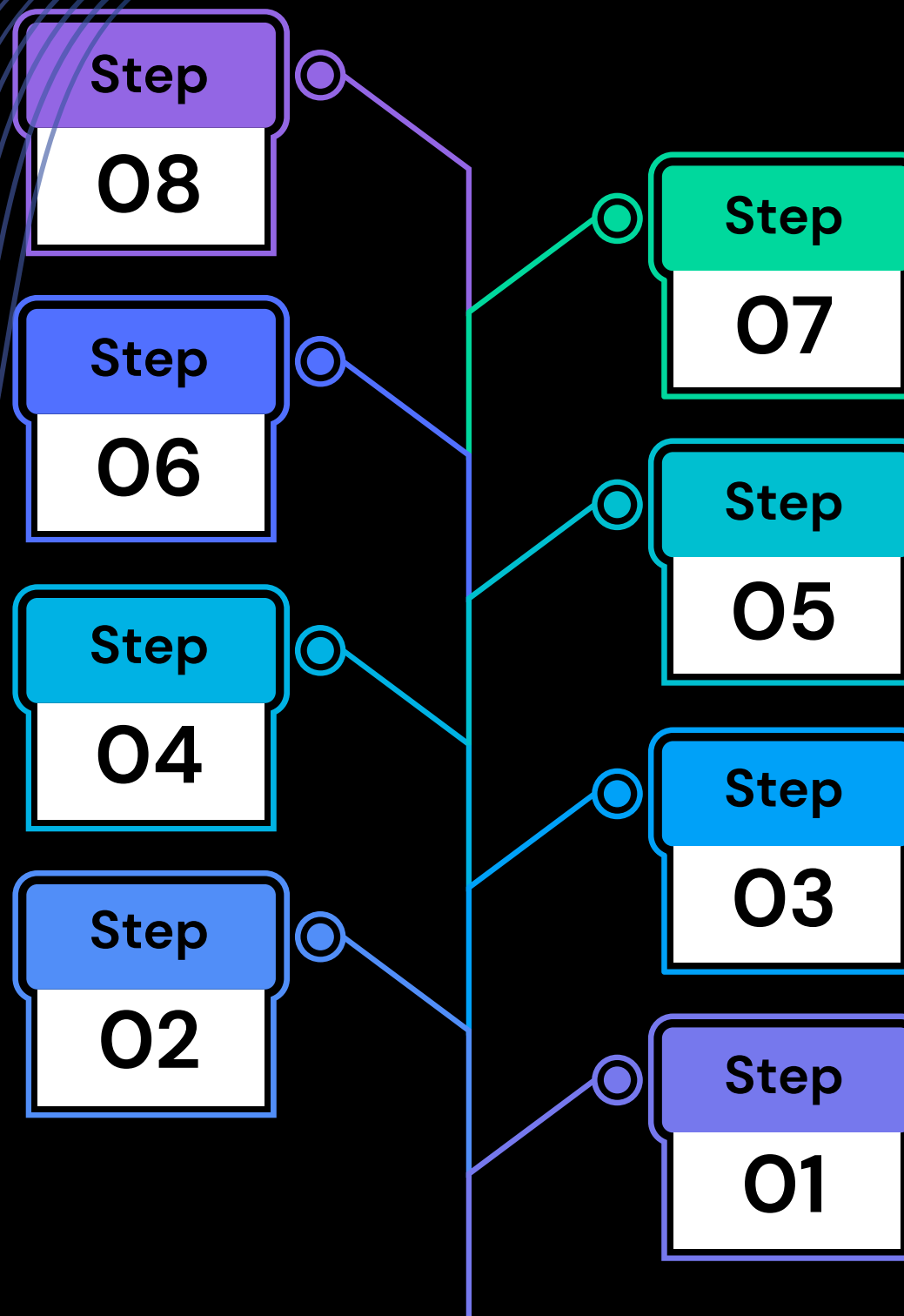


Candidate receives transparent feedback

Recruiter reviews AI insights

AI analyzes resumes

Upload resume

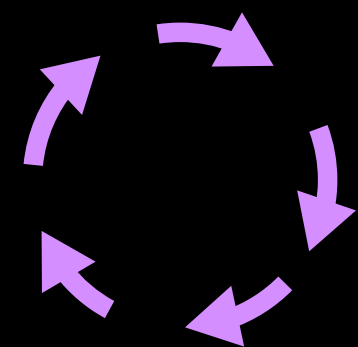


Decision recorded on blockchain

Candidates ranked automatically

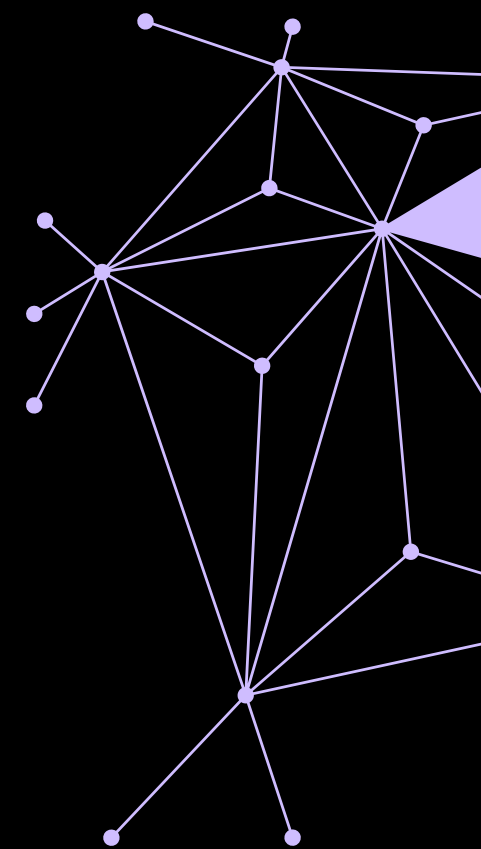
Recruiter posts job

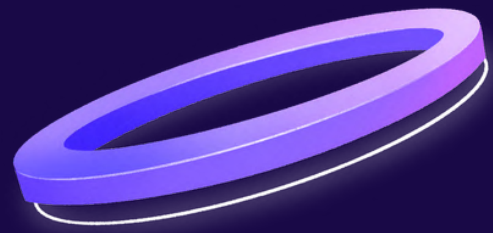
Candidate registers



System Workflow

End-to-End Hiring Process





DEMO

Live System Demonstration

1. Candidate resume upload

2. AI resume analysis

3. Candidate ranking dashboard

4. Explainable AI report

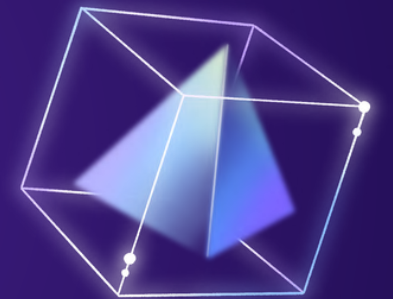
5. Blockchain verification badge



Benefits

➤ For recruiters

- 10× faster screening
- Data-driven hiring
- Bias-reduced candidate selection



➤ For candidates

- Transparent feedback
- Fair evaluation
- Verifiable decisions

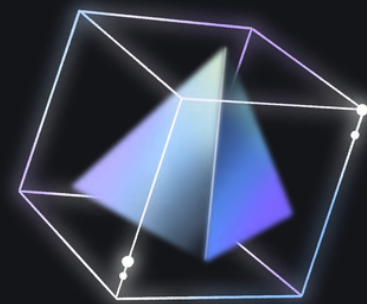


CONCLUSION



Future Improvements

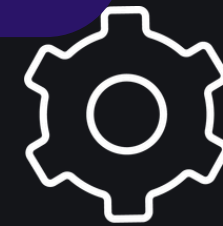
- Multi-language resume analysis
- AI interview scoring
- Cross-company skill reputation system
- Integration with LinkedIn & ATS platforms
- On-chain hiring reputation scores



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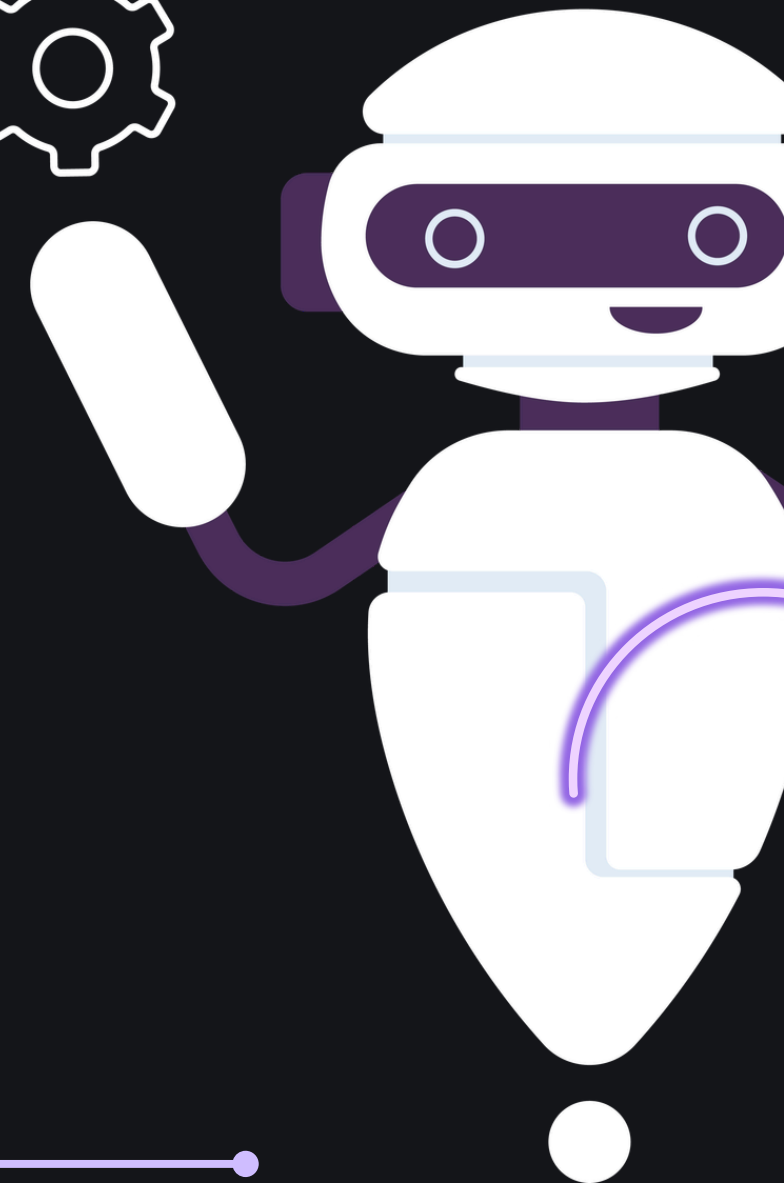
○ TalentLens AI transforms recruitment by combining:

- Artificial Intelligence
- Explainable AI
- Blockchain technology



Result:

- ✓ Fair hiring
- ✓ Transparent decisions
- ✓ Tamper-proof hiring records





THANK YOU!